

7.

Alternating current

Most of the electric power generated and used in the world is in the form of alternating current (a.c.). This is because-

- (i) Alternating voltage can easily and efficiently be converted from one value to other by means of transformers.
- (ii) The Alternating current energy can be transmitted and distributed over long distances economically without much loss of energy.

Alternating Current (a.c.)

The magnitude of alternating current (a.c.) changes continuously with time and its direction is reversed periodically.

It is represented by -

$$I = I_0 \sin \omega t$$

or $I = I_0 \cos \omega t$

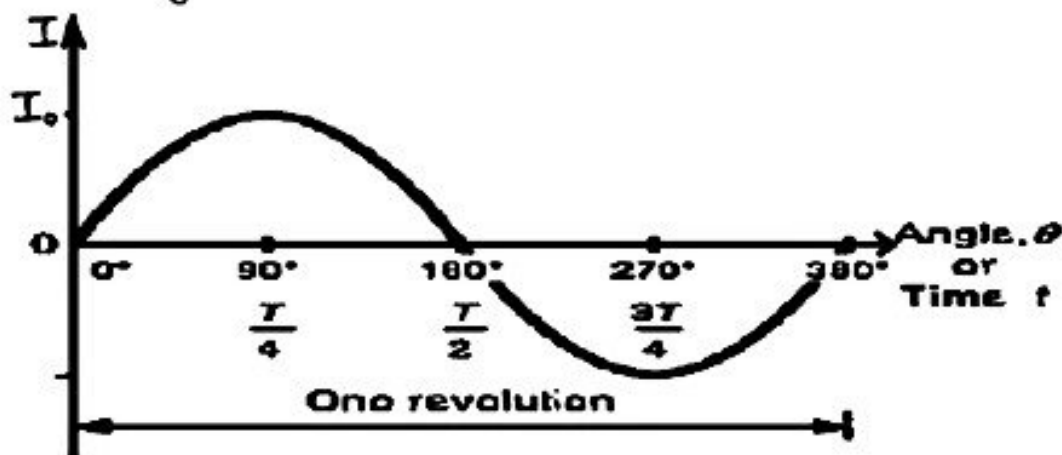
Here I is instantaneous value of current and I_0 is the peak value or maximum value of a.c. ω is called angular frequency of a.c.

Where

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$\frac{1}{T} = f \text{ (frequency)}$$

Where T is the time period of a.c. It is equal to the time taken by the a.c. to go through one complete cycle of variations.



Also f is the frequency of a.c. It is equal to the number of complete cycles of variations gone through by the a.c. in one second.

Similarly the alternating e.m.f. may be represented by-

$$E = E_0 \sin \omega t$$

or

$$E = E_0 \cos \omega t$$

In a.c. circuits, two additional circuit elements are used. These are inductor (L) and capacitor (C). Thus current and voltage in a.c. circuit are controlled by three circuit elements R , L and C .

* The voltage across a pure inductor L is given by-

$$V = L \frac{dI}{dt}$$

where dI/dt is rate of change of current.

The above equation shows that an inductor affects the voltage only when current I changes with time.

* For an ideal capacitor of capacitance C ,

$$Q = CV$$

$$\text{As } I = \frac{dQ}{dt} \quad \Rightarrow \quad I = \frac{d(CV)}{dt}$$

or

$$I = C \frac{dV}{dt}$$

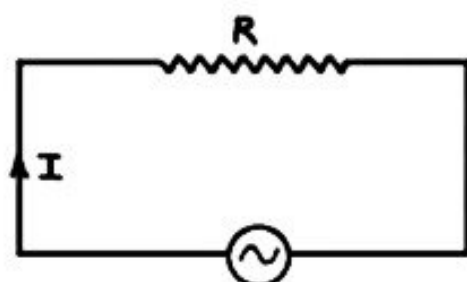
where dV/dt is rate of change of potential.

The equation shows that a capacitor affects the current only when V changes with time.

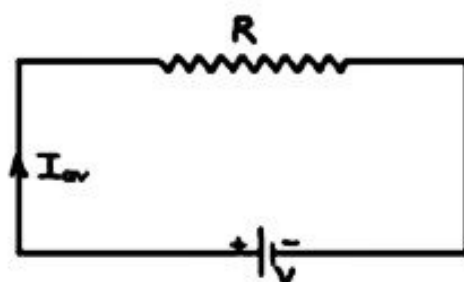
Mean or Average value of Alternating Current -

It is defined as that value of steady current which would send the same amount of charge through a circuit in the time of half cycle (i.e. $T/2$) as is sent by the a.c. through the same circuit, in the same time.

* The mean or average value of a.c. over one complete cycle is zero. So we can calculate the mean or average value of a.c. over any half cycle.



$$I = I_0 \sin \omega t$$



Let an alternating current be represented by -

$$I = I_0 \sin \omega t$$

If the strength of current is assumed to remain constant for a small time dt , then small amount of charge sent in a small time dt is -

$$dq = I dt$$

Let q be the total charge sent by a.c. in the first half cycle ($0 \rightarrow T/2$)

$$q = \int_0^{T/2} I dt \quad \text{or} \quad q = \int_0^{T/2} I_0 \sin \omega t dt$$

$$q = I_0 \left[-\frac{\cos \omega t}{\omega} \right]_0^{T/2} \quad [\omega T = 2\pi]$$

$$= -\frac{I_0}{\omega} [\cos \omega \frac{T}{2} - \cos 0^\circ]$$

$$= -\frac{I_0}{\omega} [\cos \pi - \cos 0^\circ]$$

$$Q = \frac{2 I_0}{\omega} \quad \text{--- (1)}$$

If I_{av} represents the mean or average value of a.c. over the 1st half cycle then -

$$Q = I_{av} \times \frac{T}{2} \quad \text{--- (2)}$$

from equation (1) and (2) we get -

$$I_{av} \times \frac{T}{2} = 2 \frac{I_0}{\omega} = \frac{2 I_0}{2\pi/T}$$

$$I_{av} = \frac{2 I_0}{\pi} = 0.637 I_0$$

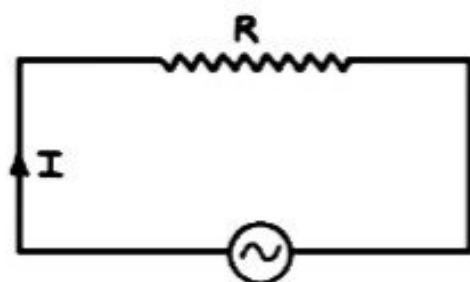
Hence mean or average value of a.c. over positive half cycle is 0.637 times the peak value of a.c., or we can say 63.7 % of the peak value.

Note -

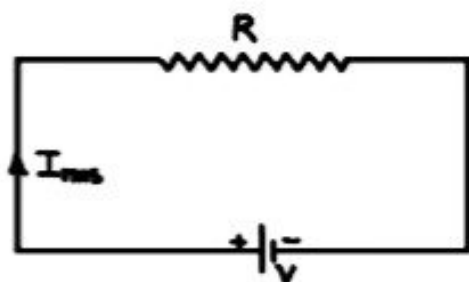
The ordinary d.c. ammeter and d.c. voltmeter cannot measure alternating current/voltage. They record zero reading, when used in a.c. circuit, because average value of alternating current/voltage over a full cycle is zero.

Root Mean Square Value of Alternating Current

It is defined as that value of steady current, which would generate the same amount of heat in a given resistance in a given time, as is done by the a.c., when passed through the same resistance for the same time.



$$I = I_0 \sin \omega t$$



The r.m.s value is also called effective value of a.c. or virtual value of a.c. It is represented by I_{rms} or I_v .

Let an alternating current is represented by -

$$\boxed{I = I_0 \sin \omega t} \quad \text{--- (1)}$$

Let this current flow through a resistance R . In a small time dt , the amount of heat produced in resistance R is -

$$dH = I^2 R dt \quad \text{--- (2)}$$

In one complete cycle (time $0 \rightarrow T$), the total amount of heat produced in the resistance R is -

$$H = \int_0^T I^2 R dt$$

$$\text{or } H = \int_0^T (I_0^2 \sin^2 \omega t) R dt$$

$$H = I_0^2 R \int_0^T \frac{(1 - \cos 2\omega t)}{2} dt$$

$$H = \frac{I_0^2 R}{2} \left[\int_0^T 1 \cdot dt - \int_0^T \cos 2\omega t dt \right]$$

$$H = \frac{I_0^2 R}{2} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T$$

$$H = \frac{I_0^2 R}{2} \left[T - \frac{\sin 2\omega T}{2\omega} \right] \quad (\because \omega T = 2\pi)$$

$$\text{So } H = \frac{I_0^2 R T}{2} \quad \text{--- (3)}$$

If r.m.s. value or virtual value of a.c. is represented by I_{rms} , then, the amount of heat produced in the same resistance R in the same time T would be -

$$H = I_{rms}^2 R T \quad \text{--- (4)}$$

So from eq. (3) and (4) -

$$I_{rms}^2 RT = \frac{I_0^2 RT}{2}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

Hence the rms. value or virtual value or effective value of a.c. is 0.707 times the peak value of a.c. or we can say 70.7 % of the peak value of a.c.

Note -

Alternating current / voltage are measured by a.c. ammeter / voltmeter respectively. These instruments are called hot wire instruments and they measure only virtual or rms. values of alternating currents / voltages.

Q. A light bulb is rated 150 W for 220 V AC supply of 60 Hz. Calculate (a) the resistance of the bulb (b) the rms current through the bulb.

Sol. Given that - $P = 150 \text{ W}$, $V = 220 \text{ V}$

(a) Resistance of the bulb $R = \frac{V^2}{P}$

$$R = \frac{220 \times 220}{150} = 322.7 \Omega$$

$$(b) I_{rms} = \frac{V_{rms}}{R} = \frac{220}{322.7} = 0.68 \text{ A}$$

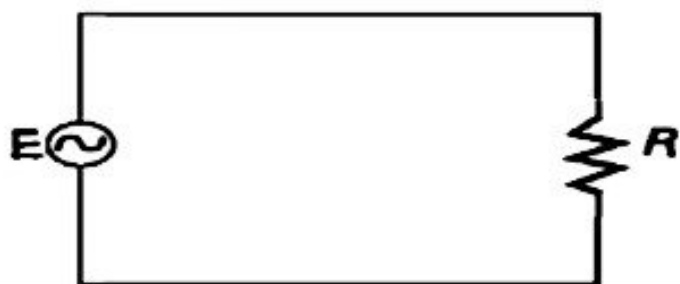
A.C. Circuit Containing Resistance Only

Let a source of alternating e.m.f. is connected to a pure resistance R .

Suppose the alternating e.m.f. supplied is represented by -

$$E = E_0 \sin \omega t$$

— (1)



Let I be the current in the circuit at any instant t . The potential difference developed across R will be IR .

This must be equal to e.m.f. applied at that instant -

$$IR = E = E_0 \sin \omega t$$

or

$$I = \frac{E_0}{R} \sin \omega t = I_0 \sin \omega t$$

— (2)

where $I_0 = E_0/R$ (Maximum value of current)

* The behaviour of R in d.c. and a.c. circuits is the same. R can reduce a.c. as well as d.c. equally effectively.

By comparing equation (1) and (2) we can see that in an a.c. circuit containing R only, the voltage and current are in the same phase.

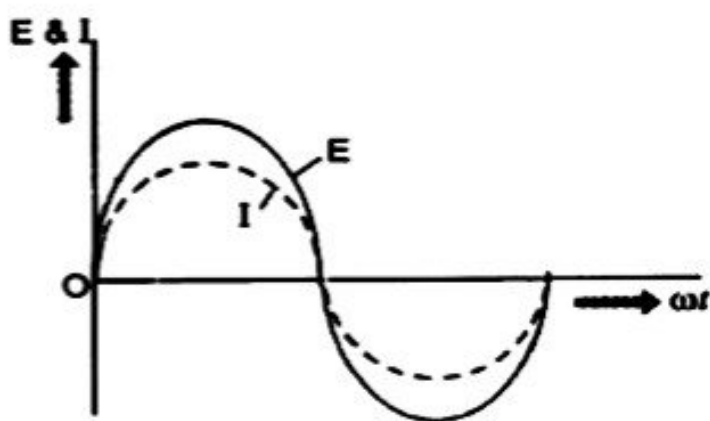
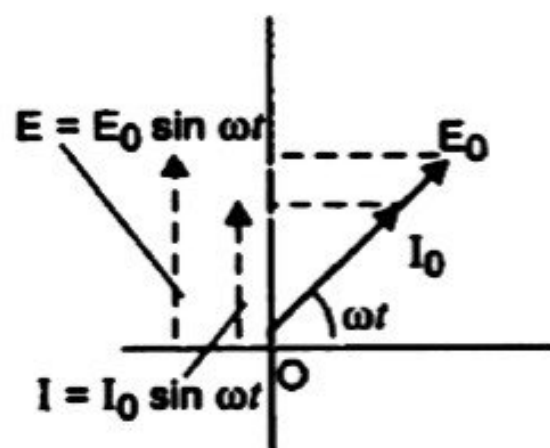
Phasor Diagram -

A rotating vector that represents a quantity varying sinusoidally with time is called a phasor.

(i) In phasor diagram, peak values of alternating current (I_0) and alternating e.m.f. (E_0) are represented by arrows called phasors.

(ii) They are inclined to horizontal axis at $\angle \omega t$ and rotate in the anticlockwise direction.

The length of the arrow represents the maximum value of the quantity i.e. I_0 and E_0 .



(iii) The projection of the arrow on any axis represents the instantaneous value of the quantity.

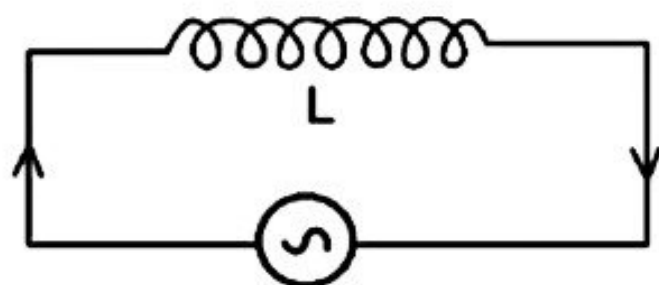
(iv) The phasor difference between the two alternating quantities is represented by the angle between the two vectors $\vec{E}_0$ and $\vec{I}_0$.

(v) In the a.c. circuit containing R only, current and voltage are in the same phase i.e. the phase angle between alternating voltage and current through R is zero.

A.C. Circuit Containing Inductance only

Let a source of e.m.f. is connected to a circuit containing a pure inductance L only. Suppose the alternating e.m.f. supplied is given by.

$$E = E_0 \sin \omega t$$



$$E = E_0 \sin \omega t$$

If dI/dt is the rate of change of current through L at any instant, then induced e.m.f. in the inductor at the same instant is-

$$e = -L \frac{dI}{dt}$$

The negative sign indicates that induced e.m.f. oppose the change of current.

To maintain the flow of current in this circuit, applied voltage must be equal and opposite to the induced voltage.

$$E = -(-L \frac{dI}{dt}) = E_0 \sin \omega t$$

$$\text{or } dI = \frac{E_0}{L} \sin \omega t dt$$

By integrating both sides we get -

$$\int dI = \frac{E_0}{L} \int \sin \omega t dt$$

$$I = \frac{E_0}{L} \left(-\frac{\cos \omega t}{\omega} \right) = -\frac{E_0}{\omega L} \cos \omega t$$

$$I = -\frac{E_0}{\omega L} \sin \left(\frac{\pi}{2} - \omega t \right)$$

$$\text{or } I = \frac{E_0}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

The current (I) will be maximum (I_0), when $\sin(\omega t - \pi/2) = \text{maximum} = 1$

So

$$I_0 = \frac{E_0}{\omega L}$$

Hence

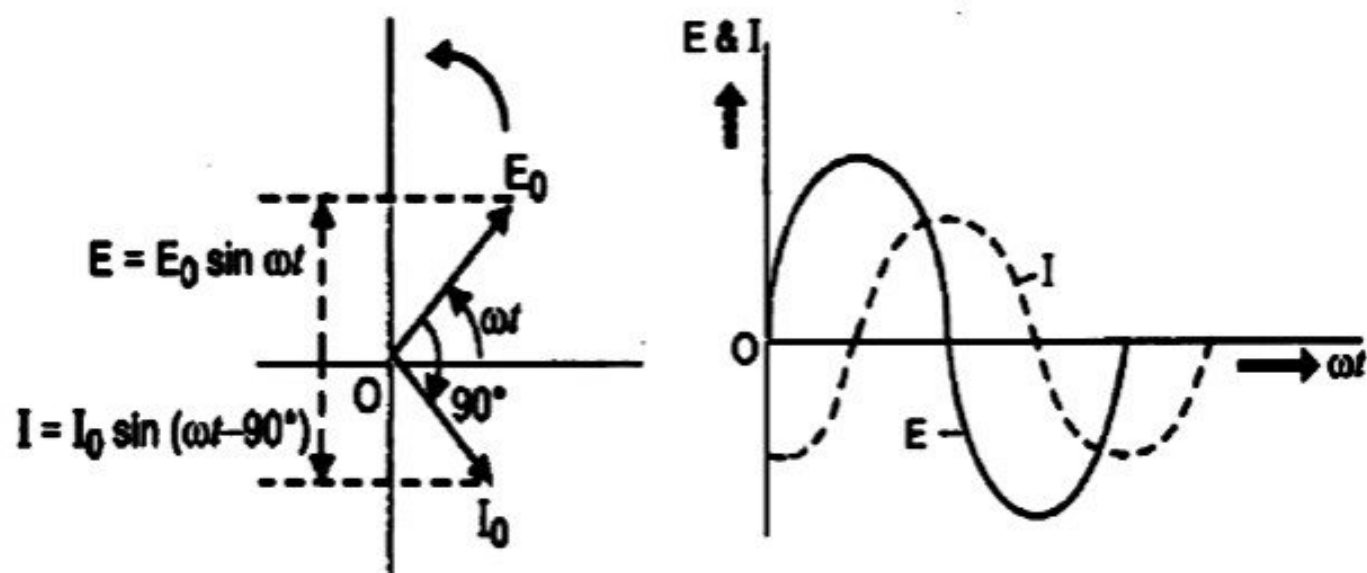
$$I = I_0 \sin(\omega t - \pi/2)$$

So we find that in an a.c. circuit containing L only, alternating current I lags behind the alternating voltage E by a phase angle of 90° . or we can say one fourth of a period.

Phasor Diagram -

In the phasor diagram of a.c. circuit containing L current lags behind the e.m.f. by 90° , so the phasor representing I_0 is turned clockwise through 90° from the direction of E_0 .

The projections of these phasors on y axis give instantaneous values E and I .



Here ωL represents the effective resistance offered by inductance L . This is called inductance reactance and is denoted by X_L .

So

$$X_L = \omega L = 2\pi f L$$

$f \rightarrow$ frequency of a.c. supply

* In dc circuits $f = 0$ so $X_L = 0$

so that a pure inductance offers zero resistance to d.c. It means a pure inductor cannot reduce d.c.

SI unit of inductive reactance is ohm. (Ω)

$$X_L = \omega L = \frac{\text{henry}}{\text{sec.}} = \frac{\text{Volt}}{\text{sec.} \times \text{amp./sec.}} = \text{ohm.} (\Omega)$$

Q. A 44 mH inductor is connected to 220 V, 50 Hz ac. supply. Determine the r.m.s. value of current in the circuit.

Sol. Given that $L = 44 \text{ mH} = 44 \times 10^{-3} \text{ H}$
 $E_{\text{rms}} = 220 \text{ V}$, $f = 50 \text{ Hz}$

$$\begin{aligned} \text{So } X_L &= \omega L = 2\pi fL \\ X_L &= 2 \times 3.14 \times 50 \times 44 \times 10^{-3} = 13.83 \text{ ohm} (\Omega) \end{aligned}$$

$$I_{\text{rms}} = \frac{E_{\text{rms}}}{X_L} = \frac{220}{13.83} = 15.9 \text{ A}$$